

Mixing and Flow Behavior of Pea Protein Isolate: Implications for Feeding System Design

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Mixing and Flow Behaviour of Pea Protein Isolate: Implications for Feeding System Design

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by

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based on research carried out

under the supervision of

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Dedicated to my beloved Parents...



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April 30, 2026

Supervisor's Certificate

This is to certify that the work presented in this dissertation entitled *Mixing and Flow Behaviour of Pea Protein Isolate: Implications for Feeding System Design* by Mr. Gunakar Soren, Roll Number 122FP0880, is a record of original research carried out by him under my supervision and guidance in partial fulfillment of the requirements of the degree of *Bachelor of Technology in Food Process Engineering*. Neither this dissertation nor any part of it has been submitted for any degree or diploma to any institute or university in India or abroad.

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Declaration of Originality

I, *Gunakar Soren*, Roll Number *122FP0880*, hereby declare that this dissertation entitled "*Mixing and Flow Behaviour of Pea Protein Isolate: Implications for Feeding System Design*" represents my original work carried out as a B.Tech. student of NIT Rourkela and, to the best of my knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of NIT Rourkela or any other institution. Any contribution made to this research by others, with whom I have worked at NIT Rourkela or elsewhere, is explicitly acknowledged in the dissertation. Works of other authors cited in this dissertation have been duly acknowledged under the "Reference" section. I have also submitted my original research records to the scrutiny committee for the evaluation of my dissertation.

I am fully aware that in case of any non-compliance detected in the future, the Senate of NIT Rourkela may withdraw the degree awarded to me based on the present dissertation.

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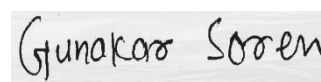
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Abstract

Efficient feeding in twin-screw extrusion systems depends on the uniformity of moisture distribution and mixing of the feed material. In this study, the combined effect of moisture content (40–60%) and mixing speed (low, medium, and high) on the mixing behavior of pea protein isolate flour was investigated using a planetary mixer. Mixing performance was evaluated using the Lacey Mixing Index (MI) by analyzing moisture distribution over time. The results showed that the mixing index increased progressively with time for all conditions, indicating improved homogeneity. Rapid mixing was observed during the initial stages, with effective mixing achieved around 3 minutes and near-uniformity attained by 9 minutes ($MI \approx 0.944$). Higher mixing speeds significantly reduced the time required to reach uniformity, while increased moisture levels slowed down mixing due to agglomeration and reduced particle mobility. At higher moisture content, excessive mixing also resulted in wall consolidation and moisture segregation. Further analysis of bulk handling properties indicated that moisture content influenced powder behavior, where moderate moisture improved particle rearrangement but higher moisture increased cohesion and resistance to flow. Overall, the study demonstrates that both mixing speed and moisture content play a critical role in determining mixing efficiency and flow behavior, and these findings can be used to optimize feeding system design in extrusion processes.

Keywords: *Pea Protein Isolate, Mixing Index, Moisture Content, Powder Flow*

CHAPTER 1

INTRODUCTION

High Moisture Meat Analogue (HMMA) is defined as a plant-based protein product that imitates the fibrous structure and textural characteristics of regular animal meat. This product is generally manufactured through the twin-screw extrusion process conducted at elevated moisture levels (40-70%). In this procedure, plant-based proteins undergo thermomechanical processing through heat, pressure, and shear. The resulting process causes the proteins to denature, align themselves, and form anisotropic and layered protein structures similar to muscle fibers. Generally, HMMA is formulated by utilizing plant protein sources, including soy protein isolate, wheat gluten, and, increasingly, pea protein isolate (PPI) (Medina-Rendón et al., 2023). Recent years have seen growing interest in PPI owing to its non-allergenic characteristics, balanced amino acid composition, and clean label appeal. Furthermore, the functional attributes of PPI include high water absorption, gelation, and emulsification capacity.

The process of loading the hydrated protein materials into the extruder is a crucial stage in the manufacture of HMMA. The effectiveness of this stage largely depends on the degree of moisture uniformity in the feed material. Non-uniform loading of the material leads to process problems including pressure pulsations, screw starvation, and feed surging, among others. This is because moisture significantly influences the powder flow properties through its effect on the inter-particle forces such as cohesion, adhesion, and friction (Zambrano et al., 2022). High moisture levels cause increased particle cohesion and stickiness, thereby limiting their mobility and flow properties, while low moisture levels cause segregation and non-uniform powder distribution. On the other hand, mixing parameters determine the uniform distribution of moisture.

Mathematical modeling helps in developing an appropriate methodological approach to understand and predict the behavior of the aforementioned systems. In particulate substances, such as wet powders, there is non-linearity in the relation of the moisture content and the flow characteristics because of the shift from the lubrication mode to the cohesive one. Parameters, such as bulk density, tapped density, Hausner's ratio, Carr's index, and the angle of repose, are often used to estimate the characteristics of flowability and packing. Linear, quadratic, exponential, and power empirical models may be applied to estimate relations between moisture

content and these features, allowing predicting the behavior of materials at various operating conditions.

However, very little information is available in the literature regarding the role played by these variables on the mixing and flow characteristics of pea protein isolate when subjected to high moisture contents. In light of this fact, the current research is conducted with the intention of filling this knowledge gap through the generation of empirical models and gaining an insight into the interplay between moisture and mixing effects.

CHAPTER 2

REVIEW OF LITERATURE

2.1. Powder Flow Theory

Powder flow refers to the movement and behavior of particles due to gravity, vibrations, shear, or handling conditions (Peleg 1977). Flow characteristics are influenced by several parameters of the particles, such as particle size, shape, weight, and surface texture. Forces that exist between particles such as cohesion, adhesion, friction, van der Waals force, electrical charges, and capillary forces due to moisture are important determinants of powder flowability. Particles that are smaller in size or uneven in texture create strong cohesive structure and therefore poor flow properties.

2.2. Moisture Influence on Powder Flow

Water is responsible for both positive and negative impacts on the behavior of powder material. Low water content facilitates the formation of thin layers of lubricant water between the particles, which minimizes the friction forces and allows the particles to move freely. With an increased level of moisture, liquid bridges between the particles are formed due to the high cohesive forces, leading to agglomeration and adhesion (Ajabshir et al., 2025).

2.3. Mixing Effects

Mixing processes greatly impact powder flow and blending efficiency. Increased mixing rate or time results in higher content homogeneity and lower segregation tendencies, especially when dealing with powder blends having different particle size or density. Too much mixing leads to particle fragmentation and hence the formation of fines, thus increasing the surface area and creating cohesion that could negatively impact flowability. Mixing causes moisture distribution on powder particle surfaces, resulting in changes in adhesive force, capillary action,

and electrostatic attraction. This may cause a powder to switch between its states of flowability and cohesion due to mixing process (Ajabshir et al., 2025).

2.4. Raw Materials: Pea Protein isolate

Pea protein isolate (PPI) is one of the common plant proteins used in high-moisture meat analogs owing to their high protein composition ($\approx 80\%$) and functional characteristics (Strauch et al., 2021; Yuliarti et al., 2021). PPI is obtained from yellow pea by milling, extraction, and drying processes. Unlike soy, PPI is allergen-free, non-GMO, and is accepted well by consumers as a clean-label ingredient owing to the presence of balanced amino acids in it (Lu et al., 2019). From the perspective of functional characteristics, PPI exhibits good water-holding capacity, gelling, and emulsifying properties, which make it suitable for extrusion processing. However, high moisture in the material causes changes in inter-particle forces, leading to increased adhesion and tackiness of particles.

2.5. Extrusion Feeding System

Twin-screw extrusion feeding systems are designed to ensure that material is conveyed from the hopper into the extruder continuously and uniformly (Rao et al., 2021).. Normally, the hopper and screw combination is adopted for feeding, whereby the screw controls the rate at which the material is delivered into the extruder barrel. The importance of steady feeding cannot be underestimated as it enables steady residence time, pressure, and shear within the extruder, thus affecting the quality of products and process efficiency (Sule et al., 2024). The behavior of the feeding system varies according to the characteristics of the material being used (Seker, 2005).. Material parameters like bulk density, moisture content, and flowability dictate the feeding process within the hopper-to-screw passage. Steady feeding promotes uniform mass flow within the process. Consequently, material characteristics play an important role in understanding feeding dynamics within the extruder.

2.6. Problems in Feeding High-Moisture Materials

The introduction of moist feeds into twin screw extrusion systems can be difficult because there is a considerable change in the behavior of materials caused by moisture. Water introduces

modifications in the particle interactions, leading to the generation of liquid bridges and capillary effects, increasing the level of cohesion and reducing flowability. Thus, there will be a tendency for materials to cluster and resist motion, making it difficult to feed evenly into the extruder (Wu et al., 2012). These changes in the properties of materials cause technical problems that include bridging, ratholing, and irregular mass flow from the hopper-screw feeder assembly. Inconsistent moisture contents exacerbate these situations since variations in the properties of materials will occur, leading to unreliable feeding and erratic process performance. There will be disruptions in the continuous feeding of materials into the extruder. Hence, there is a need to study the behavior of high moisture materials to facilitate efficient feeding. It is important to determine how moisture content affects the properties of materials and how they influence cohesion and flowability and consequently material transport and handling into the extruder.

2.7. Research Gap

There is not much work done in this area of studying the flow characteristics of wet flour systems, especially when moisture levels change. This is due to the fact that there are fewer studies done on dry powder systems where particle-particle interactions are less complicated. The introduction of moisture changes the situation dramatically by inducing capillary forces and liquid bridges between particles, making interactions more cohesive and flow properties nonlinear.

Very little research has been conducted in relation to the storage and feeding of high moisture materials made up of pea protein isolate (PPI). It is important to conduct further research to determine how the properties of these materials are affected by moisture and their subsequent handling and feeding capabilities.

2.8. Objectives

1. To determine the mixing index and flow characteristics of pea protein isolate at different moisture levels and blender rotation speeds.
2. To evaluate and model the influence of moisture content on the bulk handling properties of PPI powder

CHAPTER 3

MATERIALS AND METHODS

3.1 Materials and Instruments

The following materials and instruments were used during the study:

- Pea Protein Isolate (PPI)
- Distilled Water
- Analytical Weighing Balance
- Laboratory Mixer
- Hot Air Oven
- Petri Plates

3.2 Methodology

3.2.1 Moisture Measurement and Water Calculation

The initial moisture content of the PPI was determined by the hot air oven method. One gram of sample was weighed into a petriplate and place in a hot-air oven at 105 °C for 24 h. The weight of empty petriplate (W_1), petriplate with the sample before drying (W_2) and after drying (W_3) were recorded, and the moisture content was calculated using the formula:

$$\text{Moisture content (\% wet basis)} = \frac{W_2 - W_3}{W_2 - W_1} \times 100 \quad \dots\dots\dots(1)$$

The experiment was repeated in triplicate. Using this value, the amount of water required to adjust the PPI to the target moisture levels (40%, 50%, or 60%) was calculated using:

$$\text{Water Required (g)} = \frac{(M_2 - M_1)}{(100 - M_2)} \times W \quad \dots\dots\dots(2)$$

Where:

M_1 = initial moisture (% wet basis)

M_2 = final/target moisture (% wet basis)

W = initial weight of the material (g)

3.2.2 Mixing Procedure

The mixing index was determined by the Lacey Mixing Index method. 500 g of PPI was weighed into the container of the planetary mixer. The pre-calculated amount of water was added at the beginning (10 s) of mixing, and the mixer was run for a total of 10 minutes at the selected speed. At the end of each minute, the mixer was stopped, and 5 samples were collected from fixed positions inside the mixer as shown in the Figure.3.1 below.



Figure 3.1 Sampling points 1–5 in the mixer bowl.

3.2.3 Moisture Analysis, Mixing Index Calculation

All collected samples were dried overnight in a hot-air oven to obtain constant weight, and the final moisture content was calculated from the corresponding weight loss. Using these moisture values, the Lacey Mixing Index (MI) was computed for each minute (1–10) using equation (3). The MI values were then used to generate Mixing Index vs. Time curves for each speed and moisture level.

The Lacey Mixing Index was calculated using:

$$MI = \frac{S_0^2 - S^2}{S_0^2 - S_R^2} \dots\dots\dots(3)$$

where

S^2 = variance of the current mixture,

S_0^2 = maximum variance (complete segregation),

S_R^2 = minimum variance (perfect mixing).

Variance formula (S^2) for the five samplings at every interval was calculated using the following equation.

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \dots\dots\dots(4)$$

Where,

x_i = moisture value of each sample,

$\bar{x}$ = mean moisture value,

n = number of samples

3.2.4 Modelling of Powder Flow Properties

Objective two of this research involved determining the effect of moisture content on the bulk handling characteristics of the PPI. First, the initial moisture content of the raw material used was found by the hot air oven method. From this value, samples of 70 g were obtained for different percentages of moisture content. This was done through the calculation of the quantity of the raw material and water for the specified percentage. An illustration of how to find the sample with a specified percentage of moisture is given below. In order to prepare a 10% moisture content sample, 69.25 g of PPI and 0.75 g of water were added.

The prepared samples were blended uniformly to ensure even distribution of moisture. Thereafter, the bulk characteristics, such as bulk density, tapped density, Hausner's ratio, Carr's index, angle of repose, and coefficient of friction, were estimated for each sample. The estimated results were utilized as the basis for developing mathematical models that characterize the effect of moisture content on powder behavior.

Empirical regression methods were utilized for mathematical modeling. Moisture content was taken as the independent variable (x), whereas the corresponding flow behavior was the dependent variable (y). Five distinct mathematical models, namely linear, quadratic, exponential, power models, and logarithmic model, were employed for developing the above relationships. General expressions for the above models can be given as follows: linear model

($y = a + bx$), quadratic model ($y = a + bx + cx^2$), exponential model ($y = ae^{bx}$), power model ($y = ax^b$), logarithmic model ($y = a + b \ln(x)$). These models were chosen as they are frequently used in the field of food engineering research to define linear as well as non-linear relations between the parameters under study.

Modeling and Regression were done in Microsoft Excel where curve-fitting was done on each model. Each model was tested for its performance depending on the statistical factors like the following:

Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \dots\dots\dots(5)$$

Where:

- y_i = experimental (observed) value
- $\hat{y}_i$ = predicted value from model
- $\bar{y}$ = mean of observed values

Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} \dots\dots\dots(6)$$

Where, n = number of observations

Chi-square (χ^2)

$$\chi^2 = \sum \frac{(y_i - \hat{y}_i)^2}{\hat{y}_i} \dots\dots\dots(7)$$

The model that had the highest R^2 and smallest errors was chosen as the most suitable one to characterize the relationship between moisture content and corresponding flow property.

The choice of such models depended on the need to characterize the simple and more complicated processes taking place in the system under consideration. The fact that moisture content affected powder flow in a non-linear manner was another reason why the quadratic and exponential models were included. They helped to understand the influence of moisture content on the properties of the powder.

3.2.5 Powder Flow Property Evaluation

In order to study the effect of moisture content on the bulk handling characteristics of PPI powder, a number of flowability tests were performed. These powders were made with varying percentages of moisture content (10%, 15%, 70% wet basis). This was done by mixing the calculated quantity of distilled water in PPI to give them an equal percentage of moisture content. Flowability test of these sample materials was carried out and the data obtained were analyzed through classification of flowability in accordance with Tables 3.1 and Table 3.2 below.

The following powder flow properties were measured for all wet flour.

(a) Bulk Density

$$\text{Bulk Density} = \frac{\text{Mass of Sample}}{\text{Untapped Volume}} \quad \dots \dots \dots (8)$$

(b) Tapped Density

$$\text{Tapped Density} = \frac{\text{Mass of Sample}}{\text{Tapped Volume}} \quad \dots \dots \dots (9)$$

(c) Hausner Ratio

$$\text{Hausner Ratio} = \frac{\text{Tapped Density}}{\text{Bulk Density}} \quad \dots \dots \dots (10)$$

(d) Carr's Index (Compressibility Index)

$$\text{Carr's Index (\%)} = \left(\frac{\text{Tapped Density} - \text{Bulk Density}}{\text{Tapped Density}} \right) \times 100 \quad \dots \dots \dots (11)$$

(e) Angle of Repose

$$\theta = \tan^{-1}(r/h) \quad \dots \dots \dots (12)$$

Where:

h = height of the sample heap

r = radius of the base

Table 3.1 Classification of Flowability

Flowability/Flow Character	HR value	CI(%)
Good / Free-flowing	< 1.25	< 15
Fair to Poor / Moderate flow	1.25 – 1.40	15-25
Poor / Very poor / Cohesive	≥ 1.40	≥ 25

Table 3.2 Angle of Repose Flowability Category

θ (degrees)	Flowability
$\leq 30^\circ$	Excellent / Very free-flowing powder
31–40°	Good to Fair flow
41–45°	Passable / Moderate flow
$> 45^\circ$	Poor / Very poor flow / Cohesive powder

(f) Coefficient of friction measurement

The coefficient of friction was obtained from a coefficient of friction testing apparatus with varied contact surfaces, that is wood, steel, and glass. A fixed mass of powder was laid out on the surface being tested, after which the surface was slowly raised until the powder began sliding.

The angle where sliding happened was recorded as the angle of slide (θ). This was done with varying moisture contents on each contact surface; that is 40%, 50%, and 60% wet basis. The coefficient of friction was then calculated from the formula:

$$\mu = \tan(\theta) \quad \dots\dots\dots(13)$$

This method provides an estimate of wall friction behavior, which is important for understanding powder flow in hopper and feeder systems.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Effect of Mixing Speed on Mixing Index

The quantity of water to be used was determined and summarized in Table 4.1. The relationship between the change in mixing index with time under various speeds and moisture levels was plotted in Figure. 4.1. As shown in Figure. 4.1, the mixing index showed an increasing trend with increasing mixing time for all test conditions. However, the rate of increase in the mixing index varied significantly with respect to mixing speed. Under 40% moisture level, fast mixing showed a sharp increase in the mixing index compared to medium and low mixing speeds, where the values were near one within a relatively shorter period. Medium mixing speed showed an intermediate response, whereas low mixing speed showed a relatively slower increase in the mixing index.

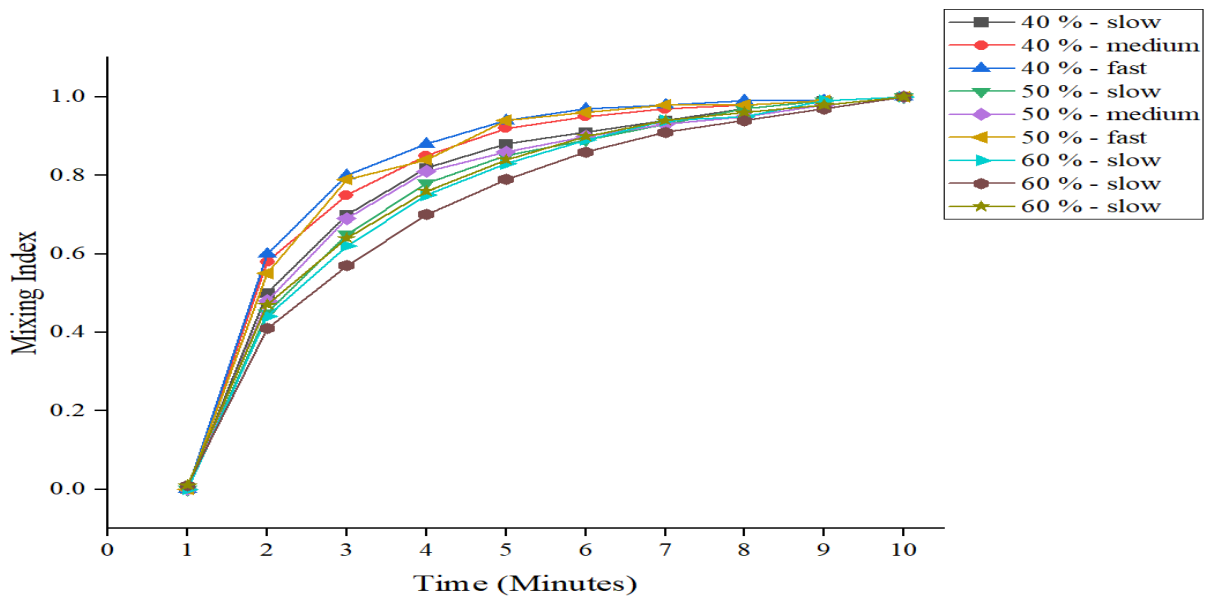


Figure. 4.1 Mixing index as a function of time at different moisture contents (40%, 50%, and 60%) and mixing speeds (slow, medium, and fast).

Table 4.1 Amount of water required to increase the moisture of PPI to target value

Target Moisture (%wb)	Initial Moisture	Sample Weight (g)	Water required (g)
40	8.88	500	259.33
50	8.88	500	411.2
60	8.88	500	639

4.2 Influence of Moisture Content

The moisture content greatly affected the mixing efficiency, as shown in Figure. 4.1 at 40% moisture content; the mixture retained its flowability, thereby making it easy for particles to move around until homogenous mixing is achieved. When the moisture content was relatively high (50% and 60% wb), the rate of change in the mixing index was relatively slow, especially in the early stages of mixing. This may be due to the increased stickiness and cohesion of the material when there was high moisture content. The existence of excess moisture causes the formation of liquid bridges, which increases the cohesion among the particles while minimizing their movement.

4.3 Mixing Behaviour Over Time

The mixing process exhibited a characteristic three-stage behavior, as clearly illustrated in Figure. 4.1

- Stage 1 (0-1 min): The values for the mixing index were low because of poor moisture distribution and inadequate particle contact.
- Stage 2 (2-5 min): The value for the mixing index rose rapidly as a result of the macro-mixing stage, which is characterized by bulk motions.
- Stage 3 (>6 min): The value of the mixing index plateaued, signifying that micro-mixing began in this stage, where diffusion and local arrangements dominated the process. In this stage, further gains would only happen very slowly.

The most interesting thing about the test was the early plateauing at the high mixing rate at 40% moisture.

4.4 Combined Effect of Mixing Speed and Moisture Content

It was found that both mixing speed and moisture content played an important role in mixing effectiveness. When the moisture content was 40%, the mixing efficiency was ranked as follows: Fast > Medium > Slow. This implies that high mixing speed is very efficient when the material has good flowability. In contrast, when the moisture content increased (50% and 60%), the difference in mixing speed became less noticeable. This is because when there is

more moisture, the material becomes more cohesive and agglomerates; therefore, it does not move effectively, thus making the mixing speed ineffective.

4.5 Mathematical Modeling of Powder Flow Properties

4.5.1 Modeling of Angle of Repose

The impact of moisture content on the angle of repose was studied using several mathematical models, such as linear, quadratic, power, exponential, and logarithmic models. The fitting results for each of the models are given in Table 4.2, whereas Figure. 4.2 shows the relation between angle of repose and moisture content.

There is a direct relationship between angle of repose and moisture content, where the former decreases with an increase in the latter, which means there is increased flowability and reduced inter-particle friction. This is because the presence of liquid film on particle surface enhances their mobility.

The quadratic model proved to be the best among all tested models, since it had the largest coefficient of determination ($R^2 = 0.965$) and smallest RMSE value (0.340), hence showing a strong non-linear correlation between two variables under study. In addition, small χ^2 value supports the use of this model.

On the other hand, the power model had the lowest coefficient of determination ($R^2 = 0.940$) and largest RMSE value, so it can be considered least suitable for modeling the relation. At the same time, linear and exponential models showed a good level of agreement with experimental data.

Table 4.2 Mathematical modelling of angle of repose as a function of moisture content

Model	Constants	R^2	RSME	χ^2
Linear model	a=50.226; b=-0.095	0.961	0.357	0.138
Quadratic model	a=50.669; b=-0.124; c=0.0003	0.965	0.34	0.125
Power model	a=58.449; b=-0.0653	0.94	0.447	0.216
Exponential model	a=50.359; b=-0.002	0.963	0.349	0.132
Logarithmic mod	a=57.349; b=-3.088	0.945	0.427	0.197

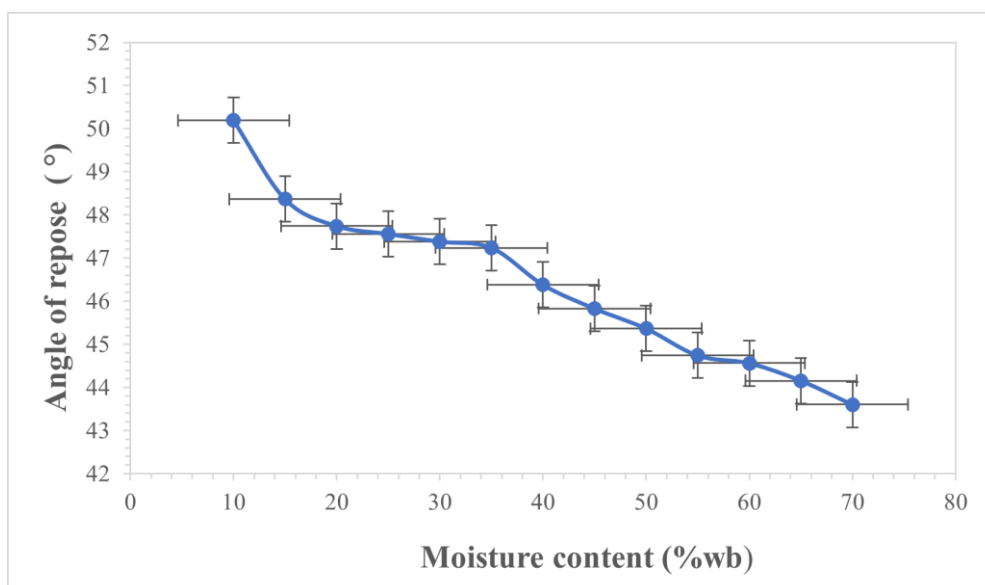


Figure. 4.2 Effect of moisture content on angle of repose

4.5.2 Modeling of Bulk Density

The relationship between moisture content and bulk density was evaluated using various mathematical models, including linear, quadratic, power, exponential, and logarithmic models, to identify the model that best describes the variation of bulk density with moisture content. The outcomes of model fitting are presented in Table 4.3, whereas the relationship between bulk density and moisture content is illustrated in Figure 4.3.

Table 4.3 Mathematical modelling of bulk density as a function of moisture content

Model	Constants	R ²	RSME	χ^2
Linear model	a=0.2493; b=-0.0018	0.9826	0.0044	2.1836 x 10 ⁻⁵
Quadratic model	a=0.2579; b=-0.0011; c=8.3273 x 10 ⁻⁶	0.9887	0.0036	1.4184 x 10 ⁻⁵
Power model	a=0.1656; b=0.1856	0.9095	0.0102	0.0001
Exponential model	a=0.3493; b=-0.0022	0.9794	0.0474	0.0024
Logarithmic model	a=0.1235; b=0.0558	0.8803	0.0118	0.0001

With an increase in moisture content, bulk density was observed to increase, showing better packing of particles and less voids. The increase in bulk density could be related to good particle rearrangement and reduced resistance between the particles, facilitated by moisture. Of all the

models tested, the quadratic model showed the closest match, with the highest R^2 value (0.9887), and lowest RMSE (0.0036) and χ^2 (1.4184×10^{-5}).

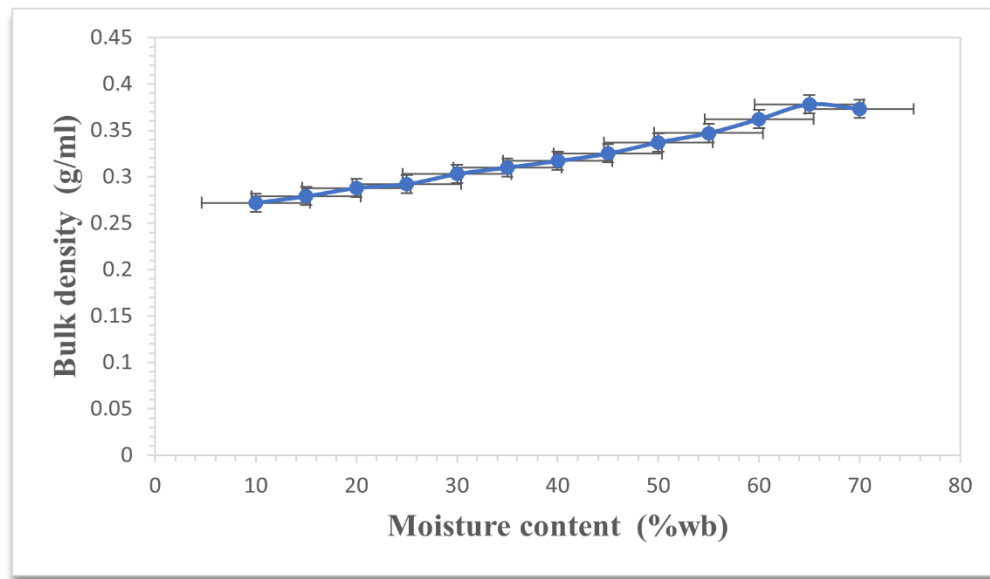


Figure. 4.3 Effect of moisture content on bulk density

Also, the linear model yielded comparable results with the actual data, although it had slightly lower prediction accuracy compared to the quadratic model. On the other hand, the exponential model had average fitting ability, while the power and logarithmic models had comparatively low R^2 values, suggesting poor fitting ability.

In summary, the findings prove that the relationship between the moisture content and the bulk density is non-linear and can be modeled using the quadratic model.

4.5.3 Modeling of Tapped Density

Moisture content effects on tapped density were studied using different mathematical models such as linear, quadratic, power, exponential, and logarithmic models. The outcome of these models is shown in Table 4.4, and the relationship between tapped density and moisture content is depicted in Figure. 4.4.

It was observed that the value of tapped density increased with an increase in moisture content, which suggests that better particle arrangement occurred when moisture content increased due to reduced void volumes between particles. This phenomenon can be explained by the fact that with an increase in moisture content, the inter-particle friction became less, resulting in better particle arrangement during tapping.

The quadratic model, compared to all other models used for the analysis, gave the best prediction results, with a high coefficient of determination ($R^2 = 0.9782$), with relatively low RMSE (0.0059) and χ^2 (3.7776×10^{-5}). This suggests that there is a very strong relationship between moisture content and tapped density. The exponential model has given very similar results ($R^2 = 0.9779$), while the linear model has a little lower R^2 value than both previously mentioned models. The power and logarithmic models have much lower values of R^2 .

The final conclusion drawn from the results is that there is a considerable effect of moisture content on tapped density since it contributes to increasing packing efficiency when exposed to mechanical agitation.

Table 4.4 Mathematical modelling of tapped density as a function of moisture content

Model	Constants	R^2	RSME	χ^2
Linear model	$a=0.4020$; $b=0.0020$	0.9747	0.0063	4.3175×10^{-5}
Quadratic model	$a=0.4163$; $b=-0.0011$; $c=1.0851 \times 10^{-5}$	0.9782	0.0059	3.7776×10^{-5}
Power model	$a=0.2940$; $b=0.1405$	0.9021	0.0124	0.0001
Exponential model	$a=0.4072$; $b=0.0043$	0.9779	0.0058	3.7704×10^{-5}
Logarithmic model	$a=0.2560$; $b=0.0647$	0.8808	0.0137	0.0002

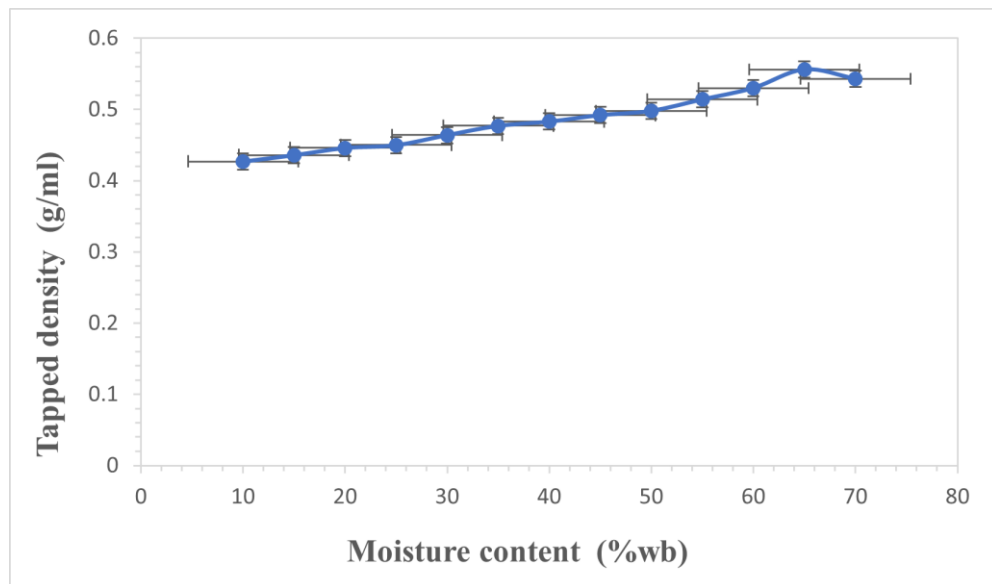


Figure. 4.4 Effect of moisture content on tapped density

4.5.4 Modeling of Hausner Ratio

The relationship between moisture content and Hausner ratio was established through various regression models, which include linear, quadratic, power, exponential, and logarithmic models. The outcomes of the regression analysis are summarized in Table 4.5, whereas the influence of moisture content on Hausner ratio is demonstrated in Figure. 4.5. The value of Hausner ratio exhibited a slight drop with the increase in moisture content, showing only minor improvements in flowability due to lower inter-particulate friction and particle rearrangements.

Among the models evaluated, the quadratic model provided the best fit, with the highest coefficient of determination ($R^2 = 0.9540$) and the lowest RMSE (0.0080) and χ^2 (7.1072×10^{-5}) values, indicating a non-linear relationship between moisture content and Hausner ratio.

The linear and exponential models also showed comparable fitting performance, with slightly lower R^2 values (0.9522 and 0.9512, respectively) and similar error values. In contrast, the power and logarithmic models exhibited significantly lower R^2 values, indicating poorer agreement with the experimental data.

The results indicate that moisture content has a limited but noticeable effect on the Hausner ratio, and the relationship is best described by a non-linear model, particularly the quadratic model.

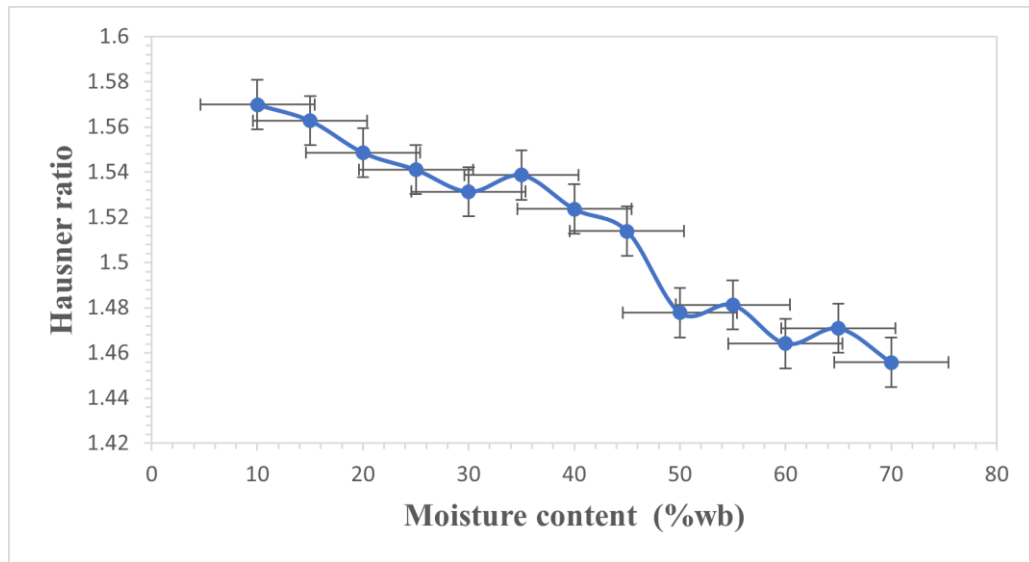


Figure 4.5 Effect of moisture content on Hausner ratio

Table 4.5 Mathematical modeling of Hausner ratio as a function of moisture content

Model	Constants	R ²	RSME	χ^2
Linear model	a=1.5926; b=-0.0019	0.9522	0.0083	7.4×10^{-5}
Quadratic model	a=1.5861; b=-0.0015; c=-5.1751 x 10 ⁻⁶	0.954	0.008	7.1072×10^{-5}
Power model	a=1.7424; b=-0.0397	0.859	0.0141	0.0002
Exponential model	a=1.5941; b=-0.0013	0.9512	0.0083	7.5417×10^{-5}
Logarithmic model	a=1.7304; b=-0.0610	0.8652	0.0138	0.0002

4.5.5 Modeling of Carr's Index

The influence of the moisture content on the Carr's Index was investigated using several types of mathematical models including linear, quadratic, power, exponential and logarithmic models. Model fitting results are presented in Table 4.6, whereas the change of Carr's Index with moisture content is demonstrated in figure. 4.6

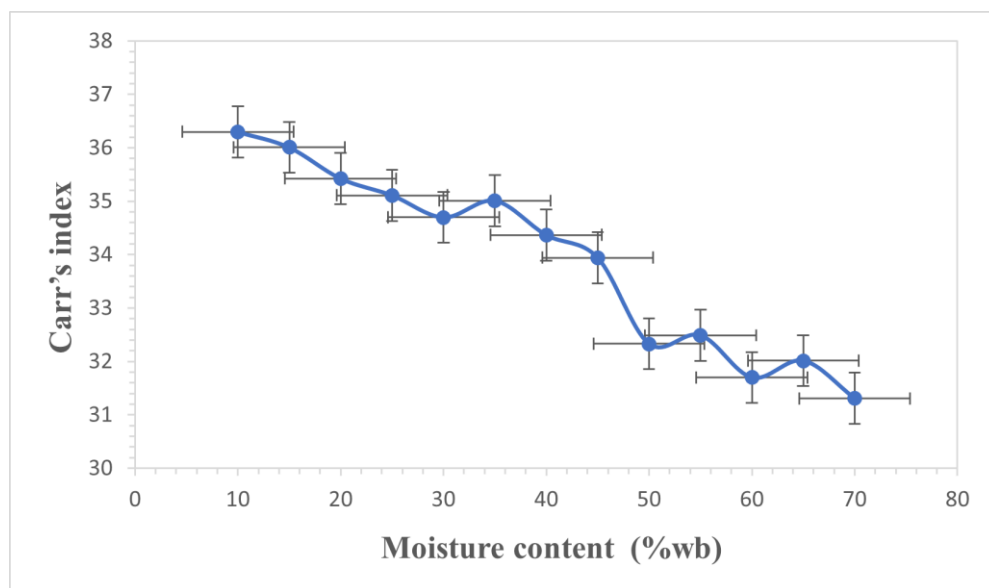
The Carr's Index decreased when moisture content increased, which means that powder became less cohesive and slightly more flowable. Such behavior may be explained by easier particles rearrangement due to moisture presence.

It was found that among all considered models, the linear model shows the best fit with moisture content. This model provides the highest value of R² (0.9493), the lowest value of RMSE (0.3731) and χ^2 (0.1508), which means that the linear relationship predominates here.

Exponential model demonstrates acceptable results with a relatively high value of R² (0.9463), however its error values are slightly higher. On the contrary, quadratic model shows poor results because its R² equals 0.9340 only. Similarly, poor results are characteristic for the power and logarithmic models. The obtained results show the strong dependence of Carr's Index from moisture content with predominant linear behavior and low compressibility.

Table 4.6 Mathematical modeling of Carr's Index as a function of moisture content

Model	Constants	R^2	RSME	χ^2
Linear model	$a=37.3212$; $b=-0.0862$	0.9493	0.3731	0.1508
Quadratic model	$a=35.9616$; $b=-0.0010$; $c=-0.0010$	0.934	0.4252	0.1959
Power model	$a=44.3650$; $b=-0.0761$	0.8418	0.6588	0.4703
Exponential model	$a=37.4792$; $b=-0.0025$	0.9463	0.3836	0.1594
Logarithmic model	$a=43.3407$; $b=-2.6613$	0.8544	0.6319	0.4327

**Figure. 4.6** Effect of moisture content on Carr's Index

4.5.6 Modeling of Coefficient of Friction

Various mathematical models were used to evaluate the influence of moisture content on the coefficient of friction, namely linear, quadratic, power, exponential, and logarithmic models. The outcome of fitting these mathematical models is illustrated in Table 4.7. Meanwhile, the relationship between the coefficient of friction and moisture content is depicted in Figure. 4.7

The coefficient of friction was found to increase as the moisture content increases. This is because of the increased resistance to flow due to high particle adhesion and cohesion. Among the fitted models, the quadratic model yielded the highest correlation coefficient ($R^2 = 0.9613$) as well as the lowest values of RMSE and χ^2 (0.0490 and 0.0026 respectively), which means there is a highly non-linear dependence between moisture content and coefficient of friction. For the exponential curve-fitting method, there was also good correspondence with the experimental results ($R^2 = 0.9401$), but with somewhat larger error values. With regard to the linear dependence, the degree

of fitting was less ($R^2 = 0.8517$). The power and logarithmic dependencies yielded even lower R^2 values.

Thus, it can be concluded that the coefficient of friction shows a significant non-linear dependence on moisture content.

Table 4.7 Mathematical modeling of coefficient of friction of steel as a function of moisture content

Model	Constants	R^2	RSME	χ^2
Linear model	$a=0.1352$; $b=0.0123$	0.8517	0.096	0.0099
Quadratic model	$a=0.4679$; $b=-0.0089$; $c=0.0002$	0.9613	0.049	0.0026
Power model	$a=0.0303$; $b=0.8223$	0.8268	0.106	0.0121
Exponential model	$a=0.2403$; $b=0.0218$	0.9401	0.0616	0.0041
Logarithmic model	$a=-0.6373$; $b=0.3565$	0.6766	0.1418	0.0217

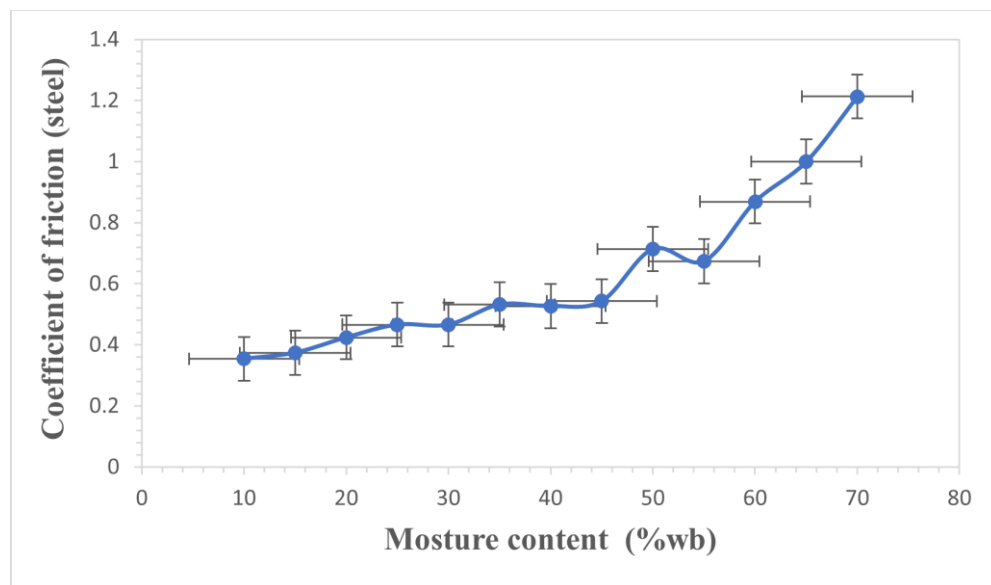


Figure. 4.7 Effect of moisture content on coefficient of friction of steel

CHAPTER 5

CONCLUSION

This research is based on the experiment which was conducted with the purpose of examining the effect of moisture and mixing conditions on the flow and mixing of PPI powder. Based on the findings concerning mixing index, the following conclusions were drawn: an increase in the speed of mixing has a positive effect on mixing, and an increase in moisture has a negative effect on mixing due to cohesion and agglomeration.

With regard to flow characteristics analysis, it may be stated that moisture has a considerable impact on flow characteristics of the examined material. For example, increasing moisture content up to an optimum level improved the flowability of the material due to the reduction in the angle of repose. An increase in moisture content also contributed to improvements in bulk density, tapped density, Carr Index, and Hausner Ratio. However, further increase in moisture content beyond the optimum level caused stickiness and increased inter-particle cohesion, thereby reducing the flowability of the material. Mathematical modeling indicated that in most cases, the relationship between the flow characteristics and moisture was not linear, and the best fit was found using quadratic correlation. The mathematical modeling assisted in capturing the difference in flow characteristics between packing-controlled to cohesive-controlled flow behavior.

According to the results obtained, the moisture plays dual roles in the flow characteristics of the powder material. The moisture improves the flow characteristics of the powder materials at medium moisture values; however, the flow characteristics become poor when moisture is very high. Thus, the above information will be useful in designing hoppers and feeders. The moisture control, mixing, steeping of the hopper, smooth walls, and large openings should be considered during the feeding process.

REFERENCES

- Ajabshir, S. Z., Barletta, D., & Poletto, M. (2025). The effect of process conditions on powder flow properties for slow flow regimes. *KONA Powder and Particle Journal*, 42, 100–120. <https://doi.org/10.14356/kona.2025006>
- Ajabshir, S. Z., Hare, C., Barletta, D., & Poletto, M. (2025). Discrete Element Method study on spreading behaviour of non-spherical polymeric powder in powder bed fusion with blade and roller spreaders. *Chemical Engineering Transactions*, 117, 787–792. <https://doi.org/10.3303/CET25117132>
- Lu, Z. X., He, J. F., Zhang, Y. C., & Bing, D. J. (2019). Composition and physicochemical properties of pea protein and its application in functional foods. *Critical Reviews in Food Science and Nutrition*, 59(15), 2593–2605. <https://doi.org/10.1080/10408398.2019.1651248>
- Medina-Rendon, E., Beltran-Medina, E., Guatemala-Morales, G., Padilla-Camberos, E., Corona-González, R., Mondragón-Cortez, P., & Arriola-Guevara, E. (2023). Optimization of extrusion conditions for an extruded food enriched with mango by-products (*Mangifera indica* var. Tommy Atkins) via response surface methodology. *Processes*, 11(11), 3182. <https://doi.org/10.3390/pr11113182>
- Owasit, A., Tripathi, S., Davé, R., & Young, J. (2025). Predicting powder blend flowability from individual constituent properties using machine learning. *Pharmaceutical Research*, 42(4), 665–683. <https://doi.org/10.1007/s11095-025-03855-x>
- Peleg, M. (1977). Flowability of food powders and methods for its evaluation—A review. *Journal of Food Process Engineering*, 1(4), 303–328. <https://onlinelibrary.wiley.com/doi/10.1111/j.1745-4530.1977.tb00188.x>
- Rao, R. R., Pandey, A., Hegde, A. R., Kulkarni, V. I., Chincholi, C., Rao, V., & Mutalik, S. (2021). Metamorphosis of twin screw extruder-based granulation technology: Applications focusing on its impact on conventional granulation technology. *AAPS PharmSciTech*, 23(1), 24. <https://doi.org/10.1208/s12249-021-02173-w>
- Seker, M. (2005). Residence time distributions of starch with high moisture content in a single-screw extruder. *Journal of Food Engineering*, 67(3), 317–324. <https://doi.org/10.1016/j.jfoodeng.2004.04.034>
- Strauch, R. C., & Lila, M. A. (2021). Pea protein isolate characteristics modulate functional properties of pea protein–cranberry polyphenol particles. *Food Science & Nutrition*, 9(6), 3740–3751. <https://doi.org/10.1002/fsn3.2335>
- Sule, S., Okafor, G. I., Momoh, O. C., Gbaa, S. T., & Amonyeze, A. O. (2024). Applications of food extrusion technology. *MOJ Food Processing & Technology*, 12(1), 74–84. <https://doi.org/10.15406/mojfpt.2024.12.00301>
- Tentu, R. D., Dabir, S., & Ishwarya, S. P. (2023). Technological review of high-moisture extrusion for creating whole-cut plant-based meat: An overview of the state of highmoisture extrusion technology and the corresponding state of ingredient development, technical and sensory

analysis, and infrastructure in India (White Paper). *Good Food Institute India*. <https://gfi-india.org/wp-content/uploads/2023/07/TechnologicalReview.pdf>

Wu, C. Y., Armstrong, B., & Vlachos, N. (2012). Characterization of powder flowability for die filling. *Particulate Science and Technology*, 30(4), 378– 389. <https://www.tandfonline.com/doi/full/10.1080/02726351.2011.588302>

Yuliarti, O., Kovis, T. J. K., & Yi, N. J. (2021). Structuring the meat analogue by using plant-based derived composites. *Journal of Food Engineering*, 288, 110138. <https://doi.org/10.1016/j.jfoodeng.2020.110138>

Zambrano, Y., Contardo, I., Moreno, M. C., & Bouchon, P. (2022). Effect of extrusion temperature and feed moisture content on the microstructural properties of rice-flour pellets and their impact on the expanded product. *Foods*, 11(2), 198 <https://www.mdpi.com/2304-8158/11/2/198>